```sql
create database company;
use company;

create table Departments(
dept_id INT PRIMARY KEY,
dept_name VARCHAR(50) NOT NULL,
);
create table Locations(
location_id int primary key,
location_name varchar(50) not null
);
create table Job_Roles(
role_id INT primary key,
role_name varchar(50) not null,
min_salary decimal(10,2),
max_salary decimal(10,2)
);


create table Employees(
emp_id INT PRIMARY KEY,
emp_name varchar(100) not null,
gender varchar(10),
dept_id INT,
role_id INT,
location_id INT,
join_date DATE,
status varchar(20) default 'Active',
foreign key (dept_id) references Departments(dept_id),
foreign key (role_id) references Job_Roles(role_id),
foreign key (location_id) references Locations(location_id)
);
create table Salaries(
salary_id int primary key,
emp_id int,
salary decimal(10,2),
effective_date date,
foreign key(emp_id) references Employees(emp_id)
);

create table Performance_Reviews(
review_id int primary key,
emp_id INT,
review_year INT,
rating decimal(2,1) check (rating between 1 and 5)
foreign key (emp_id) references Employees(emp_id)
);


BULK INSERT Departments
FROM 'C:\data\ds1.csv'
WITH (
    FIRSTROW = 1,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
```

```sql
);

BULK INSERT Locations
FROM 'C:\data\ds2.csv'
WITH (
    FIRSTROW = 1,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);
BULK INSERT Salaries
FROM 'C:\data\ds3.csv'
WITH (
    FIRSTROW = 1,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);
BULK INSERT Performance_Reviews
FROM 'C:\data\ds4.csv'
WITH (
    FIRSTROW = 1,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);
BULK INSERT Job_Roles
FROM 'C:\data\ds5.csv'
WITH (
    FIRSTROW = 1,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);
BULK INSERT Employees
FROM 'C:\data\ds6.csv'
WITH (
    FIRSTROW = 1,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '\n',
    TABLOCK
);
select * from Departments;
select * from locations;
select * from Salaries;
select * from Performance_Reviews;
select * from Job_Roles;
select * from Employees;
--top 10 highest performing employees
SELECT TOP 10
    e.emp_id,
    e.emp_name,
    d.dept_name,
    AVG(pr.rating) AS avg_rating
FROM Employees e
JOIN Performance_Reviews pr
```

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    ON e.emp_id = pr.emp_id
JOIN Departments d
    ON e.dept_id = d.dept_id
GROUP BY e.emp_id, e.emp_name, d.dept_name
ORDER BY avg_rating DESC;
--1.average rating per department
SELECT
    d.dept_name,
    AVG(pr.rating) AS avg_rating
FROM Performance_Reviews pr
JOIN Employees e
    ON pr.emp_id = e.emp_id
JOIN Departments d
    ON e.dept_id = d.dept_id
GROUP BY d.dept_name
ORDER BY avg_rating DESC;
--2.employees eligible for promotion (rating>4 and 2+years)
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name,
    pr.rating,
    e.join_date,
    DATEDIFF(YEAR, e.join_date, GETDATE()) AS years_of_service
FROM Employees e
JOIN Performance_Reviews pr
    ON e.emp_id = pr.emp_id
JOIN Departments d
    ON e.dept_id = d.dept_id
WHERE pr.review_year = (
        SELECT MAX(pr2.review_year)
        FROM Performance_Reviews pr2
        WHERE pr2.emp_id = e.emp_id
    )
AND pr.rating > 4
AND DATEDIFF(YEAR, e.join_date, GETDATE()) >= 2
ORDER BY pr.rating DESC;
--3.performance trend over years
SELECT
    review_year,
    ROUND(AVG(rating), 2) AS avg_rating
FROM Performance_Reviews
GROUP BY review_year
ORDER BY review_year;
--4.salary distrbution per department
SELECT
    d.dept_name,
    COUNT(e.emp_id) AS total_employees,
    SUM(s.salary) AS total_salary,
    AVG(s.salary) AS avg_salary,
    MIN(s.salary) AS min_salary,
    MAX(s.salary) AS max_salary
FROM Salaries s
JOIN Employees e ON s.emp_id = e.emp_id
JOIN Departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name
```

```sql
ORDER BY avg_salary DESC;
--5.highest and lowest salary per role
select role_name, min_salary, max_salary
from Job_Roles;
--6.salary vs performance correlation
SELECT
    e.emp_name,
    s.salary,
    pr.rating
FROM Employees e
JOIN Salaries s ON e.emp_id = s.emp_id
JOIN Performance_Reviews pr ON e.emp_id = pr.emp_id;
--7.employees below department average salary
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name,
    s.salary
FROM Employees e
JOIN Salaries s
    ON e.emp_id = s.emp_id
JOIN Departments d
    ON e.dept_id = d.dept_id
WHERE s.salary < (
    SELECT AVG(s2.salary)
    FROM Employees e2
    JOIN Salaries s2
        ON e2.emp_id = s2.emp_id
    WHERE e2.dept_id = e.dept_id
);
--8.hiring trend per year
SELECT
    YEAR(join_date) AS hiring_year,
    COUNT(*) AS total_hires
FROM Employees
GROUP BY YEAR(join_date)
ORDER BY hiring_year;
--9.headcount rate by department
SELECT
    d.dept_name,
    COUNT(e.emp_id) AS headcount
FROM Departments d
LEFT JOIN Employees e
    ON d.dept_id = e.dept_id
GROUP BY d.dept_name
ORDER BY headcount DESC;
--attrition rate by department

--12.employees overdue for promotion
SELECT e.emp_id, e.emp_name
FROM Employees e
JOIN Performance_Reviews p
    ON e.emp_id = p.emp_id
WHERE p.rating >= 4
AND DATEDIFF(YEAR, e.join_date, GETDATE()) >= 3;
--13.high salary but low performance
```

```sql
SELECT
    e.emp_id,
    e.emp_name,
    s.salary,
    pr.rating
FROM Employees e
JOIN Salaries s
    ON e.emp_id = s.emp_id
JOIN Performance_Reviews pr
    ON e.emp_id = pr.emp_id
WHERE s.salary > (
        SELECT AVG(s2.salary)
        FROM Salaries s2
)
AND pr.rating < 3;
--14.most stable department
SELECT TOP 1
    d.dept_name,
    COUNT(e.emp_id) AS attrition_count
FROM Employees e
JOIN Departments d
    ON e.dept_id = d.dept_id
WHERE e.status <> 'Active'
GROUP BY d.dept_name
ORDER BY attrition_count ASC;
--16.location wise salary comparison
SELECT
    l.location_name,
    AVG(s.salary) AS avg_salary
FROM Employees e
JOIN Salaries s
    ON e.emp_id = s.emp_id
JOIN Locations l
    ON e.location_id = l.location_id
GROUP BY l.location_name
ORDER BY avg_salary DESC;
--17.gender diversity ratio
SELECT
    gender,
    COUNT(*) AS total_count,
    ROUND(
        COUNT(*) * 100.0 / (SELECT COUNT(*) FROM Employees),
        3
    ) AS percentage
FROM Employees
GROUP BY gender;
--18.median salary calculation
SELECT AVG(salary) AS median_salary
FROM (
    SELECT
        salary,
        ROW_NUMBER() OVER (ORDER BY salary) AS row_num,
        COUNT(*) OVER () AS total_count
    FROM Salaries
) AS temp
WHERE row_num IN (
```

```sql
        (total_count + 1) / 2,
        (total_count + 2) / 2
);
--19.Employees with multiple promotions
SELECT
    emp_id,
    COUNT(*) AS promotion_count
FROM Salaries
GROUP BY emp_id
HAVING COUNT(*) > 1;
--20.Identify leadership pipeline candidates
SELECT
    e.emp_id,
    e.emp_name,
    pr.rating
FROM Employees e
JOIN Performance_Reviews pr
    ON e.emp_id = pr.emp_id
WHERE pr.rating >= 4
AND DATEDIFF(YEAR, e.join_date, GETDATE()) >= 3;
```

Output

Tables

	dept_id	dept_name
1	1	Engineering
2	2	Human Resources
3	3	Finance
4	4	Marketing
5	5	Operations
6	6	IT Support
7	7	Sales
8	8	Research and D...

	location_id	location_name
1	1	Bangalore
2	2	Hyderabad
3	3	Mumbai
4	4	Chennai
5	5	Delhi
6	6	Pune
7	7	Kolkata
8	8	Noida

	salary_id	emp_id	salary	effective_date
1	1	101	500000.00	2023-04-01
2	2	102	750000.00	2023-04-01
3	3	103	620000.00	2023-04-01
4	4	104	580000.00	2023-04-01
5	5	105	810000.00	2023-04-01
6	6	106	470000.00	2023-04-01
7	7	107	690000.00	2023-04-01
8	8	108	720000.00	2023-04-01

	review_id	emp_id	review_year	rating
1	1	101	2023	4.5
2	2	102	2023	3.8
3	3	103	2023	4.2
4	4	104	2023	2.9
5	5	105	2023	4.7
6	6	106	2024	3.5
7	7	107	2024	4.0
8	8	108	2024	3.2

	role_id	role_name	min_salary	max_salary
1	1	Software Engineer	400000.00	900000.00
2	2	Senior Software Engineer	800000.00	1500000.00
3	3	HR Executive	300000.00	600000.00
4	4	Project Manager	900000.00	1800000.00
5	5	Data Analyst	500000.00	1000000.00
6	6	DevOps Engineer	700000.00	1400000.00
7	7	QA Engineer	350000.00	750000.00
8	8	Business Analyst	600000.00	1200000.00

	emp_id	emp_name	gender	dept_id	role_id	location_id	join_date	status
1	101	Arjun Sharma	Male	1	1	1	2022-06-15	Active
2	102	Priya Nair	Female	2	3	2	2021-03-10	Active
3	103	Rahul Verma	Male	1	2	1	2020-09-25	Active
4	104	Sneha Iyer	Female	3	5	3	2023-01-12	Active
5	105	Vikram Singh	Male	2	4	2	2019-11-30	Active
6	106	Anjali Rao	Female	1	6	1	2022-08-05	Active
7	107	Karan Mehta	Male	4	7	3	2021-12-18	Inac...
8	108	Neha Kapoor	Female	3	8	2	2023-04-20	Active

Top 10 highest performing employees

	emp_id	emp_name	dept_name	avg_rating
1	105	Vikram Singh	Human Resources	4.700000
2	101	Arjun Sharma	Engineering	4.500000
3	103	Rahul Verma	Engineering	4.200000
4	107	Karan Mehta	Marketing	4.000000
5	102	Priya Nair	Human Resources	3.800000
6	106	Anjali Rao	Engineering	3.500000
7	108	Neha Kapoor	Finance	3.200000
8	104	Sneha Iyer	Finance	2.900000

Average rating per department

	dept_name	avg_rating
1	Human Resources	4.250000
2	Engineering	4.066666
3	Marketing	4.000000
4	Finance	3.050000

Employees eligible for promotion

	emp_id	emp_name	dept_name	rating	join_date	years_of_service
1	105	Vikram Singh	Human Resources	4.7	2019-11-30	7
2	101	Arjun Sharma	Engineering	4.5	2022-06-15	4
3	103	Rahul Verma	Engineering	4.2	2020-09-25	6

Performance trend over years

	review_year	avg_rating
1	2023	4.020000
2	2024	3.570000

Salary distribution per department

	dept_name	total_employees	total_salary	avg_salary	min_salary	max_salary
1	Human Resources	2	1560000.00	780000.000000	750000.00	810000.00
2	Marketing	1	690000.00	690000.000000	690000.00	690000.00
3	Finance	2	1300000.00	650000.000000	580000.00	720000.00
4	Engineering	3	1590000.00	530000.000000	470000.00	620000.00

Highest and lowest salary per role

	role_name	min_salary	max_salary
1	Software Engineer	400000.00	900000.00
2	Senior Software ...	800000.00	1500000...
3	HR Executive	300000.00	600000.00
4	Project Manager	900000.00	1800000...
5	Data Analyst	500000.00	1000000...
6	DevOps Engineer	700000.00	1400000...
7	QA Engineer	350000.00	750000.00
8	Business Analyst	600000.00	1200000...

Salary vs performance

	emp_name	salary	rating
1	Arjun Sharma	500000.00	4.5
2	Priya Nair	750000.00	3.8
3	Rahul Verma	620000.00	4.2
4	Sneha Iyer	580000.00	2.9
5	Vikram Singh	810000.00	4.7
6	Anjali Rao	470000.00	3.5
7	Karan Mehta	690000.00	4.0
8	Neha Kapoor	720000.00	3.2

```
Employees below department average salary
```

	emp_id	emp_name	dept_name	salary
1	101	Arjun Sharma	Engineering	500000.00
2	102	Priya Nair	Human R...	750000.00
3	104	Sneha Iyer	Finance	580000.00
4	106	Anjali Rao	Engineering	470000.00

```
Hiring trend per year
```

	hiring_year	total_hires
1	2019	1
2	2020	1
3	2021	2
4	2022	2
5	2023	2

```
Headcount by department
```

	dept_name	headcount
1	Engineering	3
2	Finance	2
3	Human R...	2
4	Marketing	1
5	Operations	0
6	Research ...	0
7	Sales	0
8	IT Support	0

```
Employees overdue for promotion
```

	emp_id	emp_name
1	101	Arjun Sharma
2	103	Rahul Verma
3	105	Vikram Singh
4	107	Karan Mehta

High salary but low performance

emp_id	emp_name	salary	rating

Most stable department

	dept_name	attrition_count
1	Marketing	1

Location wise salary comparison

	location_name	avg_salary
1	Hyderabad	760000.000000
2	Mumbai	635000.000000
3	Bangalore	530000.000000

Gender diversity ratio

	gender	total_count	percentage
1	Female	4	50.000000000000
2	Male	4	50.000000000000

Median salary

	median_salary
1	655000.000000

Employees with multiple promotions

emp_id	promotion_count

Leadership pipeline candidates

	emp_id	emp_name	rating
1	101	Arjun Sharma	4.5
2	103	Rahul Verma	4.2
3	105	Vikram Singh	4.7
4	107	Karan Mehta	4.0